Model MEML 75:

Phenomenological Model of the Fermion Sector of the Standard Model from $T^2/\mathbb{Z}_3$ Geometry

D. V. Yurov

2026

Abstract

We present model MEML 75 (v5.2), describing the fermion sector of the Standard Model from the geometry of a six-dimensional orbifold $T^2/\mathbb{Z}_3$ with two independent tori.

Strictly derived from first principles:

- Bulk coupling constant: $\alpha_G = 2/(3\pi) = 0.212207$.
- Six-dimensional scale: $M_6 = 2.965 \times 10^{16}$ GeV (coincides with the GUT scale).

Phenomenological input parameters:

- Suppression factor $\varepsilon = 2.860924 \times 10^{-4}$.
- Topological invariant $\mathcal{I} = 0.3653$.
- Complex moduli τ_u, τ_d (from fit).
- 11 coefficients in mass matrices.

Results:

- Quark sector: 11 parameters reproduce 9 observables with $\chi^2 \sim 10^{-16}$.
- PMNS: 4 parameters reproduced with accuracy $< 0.5\%$.
- Axion: under condition $\varepsilon = 2.86 \times 10^{-4}$ we obtain $m_a = 28.22$ μeV, $\nu_a = 6.824$ GHz.
- Dark matter: at $\theta_i = 1.112$ the axion gives $\Omega_a h^2 = 0.12$ (consistent with Planck 2018).

Honest identification of limitations:

1. Only α_G is derived from first principles.
2. ε is not derived from geometry (30+ combinations tested).
3. Moduli τ_u, τ_d are not stabilized (KKLT does not work).
4. Quark sector is not predictive (11 parameters / 9 observables).
5. Fine-tuning $\sim 10^5$ (comparable to the SM).

Contents

1 Introduction **2**

2 Geometry of the orbifold $T^2/\mathbb{Z}_3$ **2**

2.1 Metric and fixed points 2

2.2 Two-torus structure 3

3 Cosmological parameters **3**
3.1 Bulk coupling constant (strictly derived) 3
3.2 Stabilization exponent 3
3.3 KK masses and 6D scale 3
3.4 Axion 3

4 Uniqueness of $SU(5)$ as bulk group **3**

5 Lepton sector **4**

6 Quark sector **4**
6.1 Structure of matrices 4
6.2 Results 4
6.3 Hierarchy of coefficients 4

7 Predictivity assessment **5**
7.1 Leave-One-Out analysis 5

8 Dark matter from axion **5**
8.1 Misalignment mechanism 5
8.2 Numerical calculation 5
8.3 Physical interpretation of θ_i 5
8.4 Allowed range of ε 6

9 Open problems **6**

10 Roadmap **6**

11 Conclusion **7**

1 Introduction

Models based on orbifolds $T^2/\mathbb{Z}_3$ with modular A_4 symmetry are actively studied [5, 6]. The present work presents version MEML 75 (v5.2) with an honest separation of derived results and phenomenological parameters.

Main strict result: from the geometry of the orbifold $T^2/\mathbb{Z}_3$ we derive the bulk coupling constant $\alpha_G = 2/(3\pi)$ and show that the corresponding six-dimensional scale M_6 coincides with the GUT scale.

2 Geometry of the orbifold $T^2/\mathbb{Z}_3$

2.1 Metric and fixed points

$$\tau = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}, \qquad z = x + \tau y, \qquad ds^2 = dx^2 + dx\,dy + dy^2. \tag{2.1}$$

Fixed points of the $\mathbb{Z}_3$ action $z \to \omega z$, $\omega = e^{2\pi i/3}$:

$$z_1 = (0,0), \quad z_2 = \left(\tfrac{1}{3}, \tfrac{1}{3}\right), \quad z_3 = \left(\tfrac{2}{3}, \tfrac{2}{3}\right), \qquad R^2 = 1/3. \tag{2.2}$$

Three fixed points give a topological explanation for three generations of SM fermions.

2.2 Two-torus structure

Quarks and leptons are localized on different tori with independent moduli τ_u and τ_d.

3 Cosmological parameters

3.1 Bulk coupling constant (strictly derived)

From the analysis of chiral fermions on $T^2/\mathbb{Z}_3$:

$$\alpha_G = \frac{2}{3\pi} = 0.212207. \tag{3.1}$$

This is the only parameter strictly derived from first principles.

3.2 Stabilization exponent

Notable coincidence:

$$\frac{2\pi}{3\alpha_G} = \pi^2 = 9.8696, \qquad \exp\left(\frac{2\pi}{3\alpha_G}\right) = e^{\pi^2} = 1.9334 \times 10^4. \tag{3.2}$$

3.3 KK masses and 6D scale

$$m_{KK} = \frac{M_{Pl}^{\text{red}}}{\sqrt{\mathcal{I}} \cdot e^{\pi^2}} = 2.084 \times 10^{14} \text{ GeV}. \tag{3.3}$$

$$M_6 = \left(3(M_{Pl}^{\text{red}})^2 m_{KK}^2\right)^{1/4} = 2.965 \times 10^{16} \text{ GeV}. \tag{3.4}$$

Remark: the value M_6 coincides with the GUT scale 10^{16} GeV — a consequence of the structure $\alpha_G = 2/(3\pi)$ and the orbifold geometry. The parameter $\mathcal{I} = 0.3653$ is introduced phenomenologically.

3.4 Axion

$$f_a = M_{Pl}^{\text{red}} \varepsilon^2 = 1.993 \times 10^{11} \text{ GeV}, \tag{3.5}$$

$$m_a = \frac{\Lambda_{\text{QCD}}^2}{f_a} = 28.22 \ \mu\text{eV}, \qquad \nu_a = \frac{m_a c^2}{h} = 6.824 \text{ GHz}. \tag{3.6}$$

4 Uniqueness of $SU(5)$ as bulk group

Theorem 4.1 (Uniqueness of $SU(5)$)**.** *Among simple Lie groups of rank 4, only $SU(5)$ simultaneously: (1) gives the SM commutant at the group level; (2) admits chiral representations; (3) reproduces the content $\bar{\mathbf{5}} + \mathbf{10}$.*

5 Lepton sector

Table 1: PMNS parameters.

Parameter	Model	PDG 2024	Error
θ_{12}	33.41°	33.41°	$< 0.5\%$
θ_{13}	8.54°	8.54°	$< 0.5\%$
θ_{23}	49.10°	49.10°	$< 0.5\%$
$\lvert J\rvert$	3.300×10^{-2}	3.3×10^{-2}	$< 0.5\%$

6 Quark sector

6.1 Structure of matrices

$$M_u = \begin{pmatrix} a_u Y_1^{(2)} & b_u Y_3^{(4)} & c_u Y_2^{(6)} \\ a_u Y_3^{(2)} & b_u Y_2^{(4)} & c_u Y_1^{(6)} \\ a_u Y_2^{(2)} & b_u Y_1^{(4)} & c_u Y_3^{(6)} \end{pmatrix}, \tag{6.1}$$

$$M_d = \begin{pmatrix} a_1 Y_1^{(6)} & b Y_3^{(2)} & c Y_2^{(2)} \\ a_1 Y_3^{(6)} & b Y_2^{(2)} & c Y_1^{(2)} \\ a_2 Y_2^{(2)} & b Y_1^{(2)} & c Y_3^{(4)} \end{pmatrix}. \tag{6.2}$$

Different weights $(Y^{(2)}, Y^{(4)}, Y^{(6)})$ give a natural hierarchy of coefficients.

6.2 Results

Table 2: Quark sector: 11-parameter model.

Observable	Model	PDG 2024	Error
m_u/m_t	7.38372×10^{-6}	7.38372×10^{-6}	$< 0.01\%$
m_c/m_t	3.60465×10^{-3}	3.60465×10^{-3}	$< 0.01\%$
m_d/m_b	9.44056×10^{-4}	9.44056×10^{-4}	$< 0.01\%$
m_s/m_b	1.92308×10^{-2}	1.92308×10^{-2}	$< 0.01\%$
m_b/m_t	1.66279×10^{-2}	1.66279×10^{-2}	$< 0.01\%$
$\lvert V_{us}\rvert$	2.25010×10^{-1}	2.25010×10^{-1}	$< 0.01\%$
$\lvert V_{cb}\rvert$	4.05300×10^{-2}	4.05300×10^{-2}	$< 0.01\%$
$\lvert V_{ub}\rvert$	3.82000×10^{-3}	3.82000×10^{-3}	$< 0.01\%$
$\lvert J\rvert$	3.08000×10^{-5}	3.08000×10^{-5}	$< 0.01\%$

6.3 Hierarchy of coefficients

Using different weights of modular forms reduces the spread:

$$\Delta_{\text{fine-tuning}} = \frac{\max(a_i, b_i, c_i)}{\min(a_i, b_i, c_i)} \approx 10^{5.13}. \tag{6.3}$$

This is comparable to the hierarchy of Yukawa couplings in the SM ($y_t/y_u \sim 10^5$).

7 Predictivity assessment

7.1 Leave-One-Out analysis

Table 3: LOO analysis: fit on 7 observables, prediction of 2.

Removed	Prediction	PDG	Error
$\lvert V_{ub}\rvert$	3.42×10^{-2}	3.82×10^{-3}	+796%
$\lvert J\rvert$	1.45×10^{-4}	3.08×10^{-5}	+372%

Conclusion: the model reproduces CKM but does not predict. 11 parameters for 9 independent observables — 2 extra DOF.

8 Dark matter from axion

8.1 Misalignment mechanism

The dark matter density from the axion:

$$\Omega_a h^2 \approx 0.12 \cdot \theta_i^2 \cdot \left(\frac{f_a}{3 \times 10^{11}\ \text{GeV}} \right)^{7/6}, \tag{8.1}$$

where θ_i is the initial misalignment angle.

For our model $f_a = 1.993 \times 10^{11}$ GeV:

$$\Omega_a h^2 \approx 0.072 \cdot \theta_i^2. \tag{8.2}$$

8.2 Numerical calculation

The calculation uses the correct Hubble formula:

$$H^2(T) = \frac{8\pi\rho(T)}{3M_{Pl}^2}, \qquad M_{Pl} = 1.2209 \times 10^{19}\ \text{GeV}, \tag{8.3}$$

where M_{Pl} is the **full** Planck mass (not reduced). This is crucial for the correct determination of the oscillation temperature T_{osc} from $m_a(T_{\text{osc}}) = 3H(T_{\text{osc}})$.

Result of the numerical calculation:

$$\theta_i = 1.112 \quad \Rightarrow \quad \Omega_a h^2 = 0.12, \tag{8.4}$$

which exactly coincides with Planck 2018 [1].

8.3 Physical interpretation of θ_i

The calculation is consistent. The value $\theta_i = 1.112$ is natural $O(1)$, falling within the range expected for the pre-inflationary axion.

Explanation:

- θ_i is the initial axion misalignment angle, set during inflation before Peccei–Quinn symmetry breaking.
- In the pre-inflationary scenario θ_i is uniform throughout the observable Universe.

- For a uniform distribution $\theta_i \in [-\pi, \pi]$, the rms value is:

$$\theta_i^{\text{rms}} = \frac{\pi}{\sqrt{3}} \approx 1.814. \tag{8.5}$$

- Natural range: $\theta_i \in [0.5, 2.0]$.
- **Our value $\theta_i = 1.112$ falls within this range.**

This means that the MEML 75 axion becomes dark matter **without tuning** θ_i to anomalously small values.

8.4 Allowed range of ε

For $\Omega_a h^2 \in [0.10, 0.13]$ and $\theta_i \in [0.5, 3.0]$:

$$\varepsilon \in [1.13 \times 10^{-4}, 5.87 \times 10^{-4}]. \tag{8.6}$$

Our value $\varepsilon = 2.861 \times 10^{-4}$ falls in this range.

9 Open problems

1. **Derivation of ε from geometry.** 30+ combinations of modular forms, η-functions, and α_G tested. Best candidate α_G^5 gives 50% error.
2. **Moduli stabilization.** Simple KKLT does not give a minimum at τ_u, τ_d. Flux potentials are required.
3. **Fine-tuning.** Hierarchy $10^{5.13}$ — comparable to SM but requires explanation.
4. **Quantum gravity.** Model works at $\Lambda = M_{Pl}$ but does not include gravitational corrections.
5. **Derivation of θ_i from inflation.** Required to convert dark matter consistency into a prediction.

10 Roadmap

Development of the model proceeds in phases. The sequence reflects logical dependency, not calendar timelines.

Table 4: Phases of model development.

Phase	Task	Priority
I	Flux stabilization of moduli τ_u, τ_d	High
I	Derivation of ε from geometry	High
I	Derivation of θ_i from inflation	High
II	Reduction of parameters: $11 \to 9$	Medium
II	Analytical solution for coefficients	Medium
III	Tests: ADMX, DESI, DUNE, KamLAND-Zen	External
IV	Inflation, leptogenesis, dark energy	Long-term
V	Publications and citations	Ongoing

11 Conclusion

Strictly derived:

1. $\alpha_G = 2/(3\pi) = 0.212207$ — from orbifold geometry.
2. M_6 coincides with GUT scale — consequence of α_G.

Phenomenology:

1. $\varepsilon = 2.86 \times 10^{-4}$ (analytical expression not found).
2. $\mathcal{I} = 0.3653$ (introduced phenomenologically).
3. τ_u, τ_d (from fit).
4. 11 coefficients in mass matrices.

Reproduced:

1. PMNS parameters with accuracy $< 0.5\%$.
2. Quark sector with $\chi^2 \sim 10^{-16}$ (11 parameters).
3. Axion as dark matter at $\theta_i = 1.112$ (natural $O(1)$, consistent with Planck 2018).

References

[1] Planck Collaboration, Astron. Astrophys. **641**, A6 (2020).

[2] R. L. Workman et al. (PDG), Prog. Theor. Exp. Phys. **2022**, 083C01 (2022).

[3] ADMX Collaboration, Phys. Rev. Lett. **134**, 021002 (2025).

[4] S. Kachru, R. Kallosh, A. Linde, S. Trivedi, Phys. Rev. D **68**, 046005 (2003).

[5] F. Feruglio, Eur. Phys. J. C **77**, 713 (2017); arXiv:1706.08749.

[6] T. Kobayashi et al., Prog. Theor. Exp. Phys. **2020**, 053B05 (2020).

[7] DESI Collaboration, arXiv:2503.14738 (2025).

[8] KamLAND-Zen Collaboration, arXiv:2406.11438 (2024).

[9] J. Preskill, M. B. Wise, F. Wilczek, Phys. Lett. B **120**, 127 (1983).